PAPER_745_Crab_Nebula_Expanding_MUGE_Pulsar_Wind_Magnetic

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PAPER_745: Crab Nebula MUGE — Expanding Supernova Remnant with Pulsar Wind

Author: Daniel T. Murphy
Framework: Universal Quantum Field Superconductive Framework (UQFF)
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Abstract

The Crab Nebula (M1) is the remnant of SN 1054, powered by a central pulsar (PSR B0531+21) at the highest rotational energy loss in the known neutron star population. Its expanding ejecta (r(t) = r_0 + v_r*t) creates a time-dependent gravitational environment unlike any static system. This paper derives the Crab Nebula MUGE incorporating: the expanding radius r(t), the pulsar wind term F_wind, and the nebular magnetic energy M_mag. The UQFF framework captures both the mechanical expansion and
the magnetospheric energy injection that power the synchrotron nebula.

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1. Introduction

The Crab Nebula is a pulsar wind nebula (PWN) — one of the brightest objects in X-rays and gamma-rays. The central pulsar (P = 33 ms, ■ = 4.2x10^{-}1^3 s/s) injects ~5x10^{3}1 W of spin-down power into the nebulae through a relativistic wind. The expanding ejecta shell moves at v_r ~ 1500 km/s, with angular size growing observably over human timescales. This expanding geometry requires r(t) to appear explicitly in the MUGE denominator, making the Crab the canonical example of a time-dependent gravitational radius.

Crab Nebula parameters:
- Distance = 2.0 kpc

- Age $\approx$ 970 yr (SN 1054)
- r_0 (initial remnant radius) $\approx$ 3x10^{1}5 m
- r_current $\approx$ 5.6 ly = 5.3x10^{1}6 m
- v_r = 1500 km/s = 1.5x10^6 m/s
- M_ejecta $\approx$ 4-5 MM_sun
- B_nebula $\approx$ 10^{-}4 T (filamentary magnetic field)
- L_pulsar = 5x10^{3}1 W (spin-down luminosity)
- P_spin = 33 ms

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2. Crab Nebula MUGE

$$
\begin{aligned}
& g_Crab(r,t) = (GM)/r(t)^2 (1+H(z)t) (1-B(t)/B_crit) \\
& + (U_g1 + U_g2 + U_g3 + U_g4) \\
& + U_i \\
& + (Lambdac^2/3) \\
& + (hbar/\sqrt{DeltaxDeltap}) integral(psiHpsi dV) (2pi/t_Hubble) \\
& + rho_ejectaVg \\
& + F_wind [pulsar wind — NEW] \\
& + M_mag [magnetic energy — NEW]
\end{aligned}
$$

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3. Time-Dependent Radius r(t)

The expanding radius is the defining feature of the Crab MUGE:

$$
\begin{aligned}
& r(t) = r_0 + v_r * t \\
& r_0 = initial radius at SN explosion (m) \\
& v_r = expansion velocity = 1.5x10^6 m/s \\
& t = time since SN 1054 (s)
\end{aligned}
$$

For t = 970 yr = 3.06x10^{1}0 s:
$$
\begin{aligned}
& r(970 yr) = 3x10^{1}5 + 1.5x10^6 x 3.06x10^{1}0 \\
& r(970 yr) ~= 4.6x10^{1}6 m ~= 4.9 ly
\end{aligned}
$$

Gravitational deceleration:
$$
\begin{aligned}
& g_grav(t) = G*M_ejecta / r(t)^2 \\

& g_grav(970 yr) = 6.674x10^{-}1^1 x (4x4x1.989x10^{3}0) / (4.6x10^{1}6)^2 \\
& g_grav ~= 5x10^{-}2^2 m/s^2 (extremely weak — expansion dominates)
\end{aligned}
$$

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4. F_wind — Pulsar Wind Term

The pulsar wind carries relativistic particles outward, creating an inward reaction force equivalent to:

$$
\begin{aligned}
& F_wind = L_pulsar / (4pir(t)^2cM_ejecta) \\
& L_pulsar = 5x10^{3}1 W \\
& r(t) = current nebula radius \\
& c = 3x10^8 m/s
\end{aligned}
$$

$$
\begin{aligned}
& F_wind = 5x10^{3}1 / (4pi(4.6x10^{1}6)^23x10^84x1.989x10^{3}0) \\
& F_wind ~= 4.8x10^{-}3^0 m/s^2 (small but physically significant over 970 yr)
\end{aligned}
$$

Cumulative momentum injection over nebula lifetime:
$$
Deltap_wind = L_pulsart/c ~= 5x10^{3}1 x 3x10^{1}0 / 3x10^8 ~= 5x10^{3}3 kgm/s
$$

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5. M_mag — Magnetic Energy Term

The nebular magnetic field stores and dissipates energy, affecting dynamics:

$$
\begin{aligned}
& M_mag = B^2/(2mu_0r(t)*rho_ejecta) [magnetic pressure divided by density] \\
& B = 10^{-}4 T (filamentary field) \\
& mu_0 = 4pix10^{-}7 H/m \\
& rho_ejecta ~= 10^{-}2^2 kg/m^3 (current)
\end{aligned}
$$

$$
\begin{aligned}
& M_mag = (10^{-}4)^2 / (24pix10^{-}74.6x10^{1}6*10^{-}2^2) \\
& M_mag ~= 2.7x10^{-}1^7 m/s^2
\end{aligned}

$$

This is the dominant non-DPM-seeded term for the Crab, controlling the synchrotron brightness evolution.

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6. UQFF Gravity Components

$$
\begin{aligned}
& U_g1: Pulsar magnetic dipole \\
& \text{mu_dipole_PSR} ~= 3.8x10^{3}0 J/T (from P, ■) \\
& B_PSR = 3.8x10^8 T (surface field) \\
& U_g1 contributes at pulsar vicinity only \\
& U_g2: Superconductive aether field threading nebula \\
& B_super aligned with pulsar rotation axis \\
& U_g2 ~= 10^5 J/m^3 (strong near pulsar wind shock) \\
& U_g3: External galactic field at 2 kpc \\
& U_g3 = GM_gal/r_gal^2 \\
& U_g4: Galactic center contribution \\
& k_4rho_vac,[SCm](M_bh/d_g)e^(-alphat)cos(pit_n)
\end{aligned}
$$

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7. Temporal Evolution Summary

Time (yr)	r(t) (ly)	g_grav (m/s^2)	M_mag (m/s^2)
0	0.3	2x10^{-}1^4	10^{-}1^6
100	1.6	5x10^{-}2^1	10^{-}1^7
970	4.9	5x10^{-}2^2	2.7x10^{-}1^7
10,000	~50	~5x10^{-}2^7	~10^{-}1^9

The magnetic term M_mag remains important relative to g_grav throughout nebula evolution.

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8. Conclusion

The Crab Nebula MUGE demonstrates that time-dependent radius r(t) is essential for accurately modeling supernova remnant gravity. The F_wind pulsar injection and M_mag magnetic energy terms together dominate over classical DPM-seeded gravity within the nebula. The UQFF successfully models the transition from SN ejecta to mature PWN through its modular environmental forcing framework.

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037*
> *(AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_U_Bi_i jet*
> *modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure,
raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{G M / r_H^2}\right)$$

where:
- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37}\;\text{J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:
$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> *Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and*
> *PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific
equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \tfrac{1}{2}(\partial_\mu \phi)^2 - \lambda\bigl(\phi^2 - v_{\text{SCm}}^2\bigr)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37}\;\text{J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda}\,v_{\text{SCm}}$.

Nine-sector closure (Session 202):
$$\mathcal{L}_{9} = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} +

\mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736\;\text{GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_U_Bi_i buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_mu \phi_{\mathrm{NS}})(\partial^\mu \phi_{\mathrm{NS}}) - V(\phi_{\mathrm{NS}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac,[SCm]}} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\mathrm{NS}}) = \frac{1}{2} m^2 \phi_{\mathrm{NS}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{NS}}^4 + \kappa \cdot \rho_{\mathrm{vac,[SCm]}} \cdot \phi_{\mathrm{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2)\phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\mathrm{seed}} \xrightarrow{\text{4 forces}} F_U_Bi_i \xrightarrow{\text{sector E-L}} \delta S/\delta \phi_{\mathrm{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,[SCm]}} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,[SCm]}} \cdot \exp\!\left(-\exp\!\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.065 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$ kg/m^3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 101, \quad n_{\mathrm{channel}} = 18/26$$

Since $p_{\mathrm{DVP}} = 101$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}}' + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^4 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_U_b \cdot \left(1 - e^{-[SSq] \cdot m/M_\odot}\right) \cdot \cos\!\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\!\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.065	PASS Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 101$	PASS Resonant
BSH layers	26 harmonic terms	j = 1...26, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day^{-}1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	1.1x10^{-}5^2 m^{-}2 (UQFF vacuum term)	1.114x10^{-}5^2 m^{-}2	Planck 2018	PASS Consistent
Proton decay rate	κ = 0.0005/day -> Γ_p suppression	< 4.17x10^{-}3^5/yr	Super-K 2024	PASS Consistent
UQFF buoyancy signature	`F_U_Bi_i` unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`*) for full UQFF-SM bridge.*

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> *Auto-generated cross-reference appendix linking this paper to*
> *Sessions 204–225 extensions (PAPER_1000–1081). Added by*
> `update_corpus_crossrefs.py` *(Session 225, April 2026).*

Paper	Title
PAPER_1000	NS Merger F_U_Bi Strain Suppression & BCS Gap
PAPER_1011	GW170817 NS Merger F_U_Bi_i 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded F_U_Bi_i with S26(3)
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1047	Type Iax Supernova Buoyancy Reversal
PAPER_1050	MUGE F_U_Bi_i Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

15 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> *Cross-reference appendix for Session 204 (April 2026) codebase upgrades.*
> *Added by* `upgrade_kozima_ramanujan_appendices.py`*. For detailed derivations,*
> *see PAPER_840/851/852/855.*

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
`fneutron_s26_coupling.py`	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
`kozima_scm_cross_section.py`	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
`kozima_wstp_kernel.py`	11-symbol Wolfram export (`UQFFKozima`)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: F_neutron^SCm = N_n *sigma_n^SCm(omega)* Phi_phonon *(F_{U,Bi}/F_U - 1)*
where sigma_n^SCm(omega,n) = sigma_0 exp[-(omega-omega_SCm)^2/(2*Gamma^2)]* (1 + [SSq]*n/26)

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
`ramanujan_polylog_s26.py`	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
`s26_wstp_kernel.py`	8-symbol Wolfram export (`UQFFS26`)	S26, R26, NaiveLi, S26VDS

Core equation: S_26(z) = Li_26(z) = eta_26(z)/(1-2^{1-26}) + 2^{1-26}/(1-2^{1-26}) * Li_26(z^2)

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
`mock_theta_q26.py`	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:
- f_26(q) = Sum_{n=0}^{25} q^{n^2} / (-q;q)_n^2
- phi_26(q) = Sum_{n=0}^{25} q^{n^2} / (-q^2;q^2)_n
- psi_26(q) = Sum_{n=1}^{26} q^{n^2} / (q;q^2)_n

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
`ramanujan_pi_uqff.py`	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
`mock_theta_pi_wstp_kernel.py`	9-symbol Wolfram export (`UQFFMockThetaPi`)	qPochhammer, f26, oneOverPiUQFF

Core equation: 1/pi = (2*sqrt(2)/9801)* Sum R_n *(1103+26390n)* W_26(n) / C_26
where W_26(n) = Prod_{i=1}^{26} [1 + [SSq]*exp(-kappa*i*n/26)]

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`*,* `CondensedPhysics2.py`*,* `MAIN_{1\_CoAnQi}.cpp`*, and Wolfram kernels (*`uqff_kozima_kernel.wl`*,* `uqff_s26_kernel.wl`*,* `uqff_mock_theta_pi_kernel.wl`*).*

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